



Clinician-centered evaluation of large language model-generated discharge summaries for longer hospitalizations



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Although large language models (LLMs) have shown promise for discharge summary generation, their performance in longer hospitalizations remains less well studied. Prior evaluations of LLM-generated discharge summaries have largely involved shorter stays and have rarely examined receiving-clinician priorities or incidental finding reporting. We compared LLM-generated and human-authored discharge summaries for 60 Internal Medicine hospitalizations lasting 7 to 21 days, with paired assessment by hospitalists and primary care physicians (PCPs). Clinician reviewers preferred LLM-generated summaries for 95% of encounters and rated them higher for quality, readability, factuality and completeness. PCPs rated LLM-generated summaries as better for understanding and communicating hospital care to patients and providing follow-up care. LLM-generated summaries had fewer clinician-reviewer annotations of either factual inaccuracies or clinically relevant omissions, primarily due to fewer omissions, with no significant differences in estimated harm potential or likelihood compared with human-authored summaries. PCPs annotated more omissions and assigned higher likelihood of harm than hospitalists. Among 31 LLM-identified radiology incidental findings, 93.5% were rated factually correct and 87.1% appropriate for reporting. In this retrospective clinician evaluation, LLM-generated summaries for longer, complex hospitalizations were rated more favorably and had fewer annotated omissions than human-authored summaries, with no significant differences in estimated harm potential or likelihood.

Hospital discharge summaries are a critical component of safe transitions of care because they communicate the hospitalization course, treatment decisions, medication changes, unresolved issues, and follow-up needs to outpatient clinicians responsible for ongoing care. High-quality discharge summaries are important for patient safety and continuity of care, as incomplete or unclear documentation can contribute to medication errors, missed follow-up, and fragmented care after discharge^{1–4}. However, writing effective discharge summaries is burdensome, particularly for longer hospitalizations, where clinicians must synthesize extensive and often complex clinical information in a concise transition-of-care document^{5,6}.

Evidence on the use of large language models (LLMs) for discharge summary documentation is growing rapidly, with multiple retrospective

studies demonstrating that LLM-generated summaries achieve comparable or better overall quality to physician-written summaries. In a prior study using data from 2–5-day hospitalizations, LLM-generated summaries significantly outperformed human-written summaries in quality and readability/conciseness, with comparable performance in factuality and completeness⁷. Williams et al. found that LLM-generated summaries for 3–6-day hospitalizations outperformed physicians on conciseness and coherence, with no significant difference in overall quality⁸. Mehri et al. found no difference in total quality scores between LLM-generated and human-authored discharge summaries⁹ and Challener et al. found that LLM-generated summaries outperformed clinicians across nine quality domains, with the greatest advantage in comprehensiveness¹⁰. A recent

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prospective study by Grolleau et al. (2026), which incorporated an agentic LLM workflow into hospital course summarization at a medical center, provides even stronger support for the real-world use of LLM-generated discharge summaries. They found that physicians incorporated LLM-generated content into 57% of their final discharge notes, and that among feedback collected on 100 summaries, 88% were rated as having no harm potential. Importantly, physician burnout scores decreased despite insignificant time savings, suggesting that LLMs may reduce provider burden through cognitive offloading¹¹.

However, important gaps remain in the current literature on LLM-generated discharge summaries. First, prior studies did not stratify by hospital length of stay^{9,11}, or largely focused on shorter hospitalizations between 2 and 8 days^{7,10,12,13}, likely because of the substantial effort required for manual chart review and the technical challenges posed by LLM context window limitations. Longer hospitalizations represent a particular challenge because they often generate large volumes of fragmented, repetitive, and clinically complex documentation, making discharge summarization substantially more burdensome for clinicians than shorter admissions¹¹. However, these richer longitudinal records may also provide LLMs with more contextual information to generate higher-quality summaries. Clinicians may therefore derive the greatest benefit from systems capable of maintaining summary quality despite increasing documentation burden. Addressing this challenge also requires technical approaches that enable processing of electronic health records (EHRs) extending beyond current model context window limitations.

Second, although prior studies have shown that LLM-generated discharge summaries can achieve quality and readability comparable to or better than human-authored summaries^{7,10}, there is limited work on how these summaries are perceived by different stakeholders in the transition of care. Only one study stratified evaluations by reviewer type, and they found that primary care physicians (PCPs) and nursing home physicians rated LLM-generated summaries as significantly more concise, whereas hospitalists did not demonstrate the same preference¹². More broadly, prior literature demonstrates important differences between hospitalist and PCP priorities for discharge summaries that are highly relevant to LLM design. PCPs consistently identify diagnoses, medication changes, pending tests, and follow-up plans as critical for outpatient care, yet these elements are frequently incomplete or missing from discharge summaries in routine practice^{2,14}. Chatterton et al. found that PCPs prefer concise summaries focused on next steps, including clearly flagged incidental findings, explicit rationale for medication changes, and specific follow-up instructions, while excessive length was a source of frustration¹⁵. Although inpatient and outpatient clinicians generally agree on the core components of discharge summaries, studies suggest that hospitalists tend to prioritize detailed descriptions of the hospitalization whereas PCPs place higher priority on actionable follow-up information such as discharge medications¹⁶. Together, these findings suggest that LLM-generated discharge summaries may benefit from prompting strategies specifically designed around PCP workflow needs, such as formatting around action items and explicit highlighting of medication results, pending results and incidental findings.

Third, prior work has not evaluated LLM-based extraction of incidental findings from radiology reports in the context of discharge summary generation, which is important given the risk of missed follow-up and patient harm. Incidental findings are common and frequently lack appropriate follow-up, yet determining which findings are clinically actionable for outpatient care requires nuanced clinical judgment^{17,18}. Although prior studies have shown that LLMs can identify incidental findings from radiology reports with reasonable accuracy^{19–21}, it remains unclear how well these systems perform when integrated into discharge summaries, where excessive reporting may also contribute to information overload for outpatient clinicians.

To address these gaps, we conducted a clinician-centered evaluation of LLM-generated discharge summaries for longer hospitalizations (7–21 days) and compared them with human-authored summaries for overall performance and clinician-annotated errors. We developed a

scalable summarization pipeline capable of processing EHR documents that exceed current LLM context-window limits and incorporated a workflow for identifying and reporting radiology incidental findings. To assess stakeholder-specific perspectives, we designed prompts around PCP informational priorities and engaged both hospitalists and PCPs in role-specific and workflow-aligned evaluations. By combining technical innovation with stakeholder-centered evaluation, this study provides new evidence on the clinical utility, safety, and workflow implications of LLM-generated discharge summaries for supporting transitions of care.

Results

Of the 400 Internal Medicine inpatient encounters extracted, 385 met inclusion criteria; 42% (162/385) were female and ages ranged from 19 to 101 years (mean 72, SD $\pm$ 16). A random sample of 60 encounters was selected, stratified to match the length of stay distribution of all the encounters. 370 of the original 654 note types were retained as LLM inputs, yielding an average reduction of 55% in EHR text volume per encounter (calculated overall and averaged across encounters, not by note type). The most common note types retained after this filtering were progress notes. Length of stay ranged from 7 to 21 days along the distribution shown in Supplementary Table 5. Length-of-stay values over 14 were intentionally oversampled to better represent longer stays in the evaluation.

Performance and error characteristics of LLM-generated and human-authored summaries

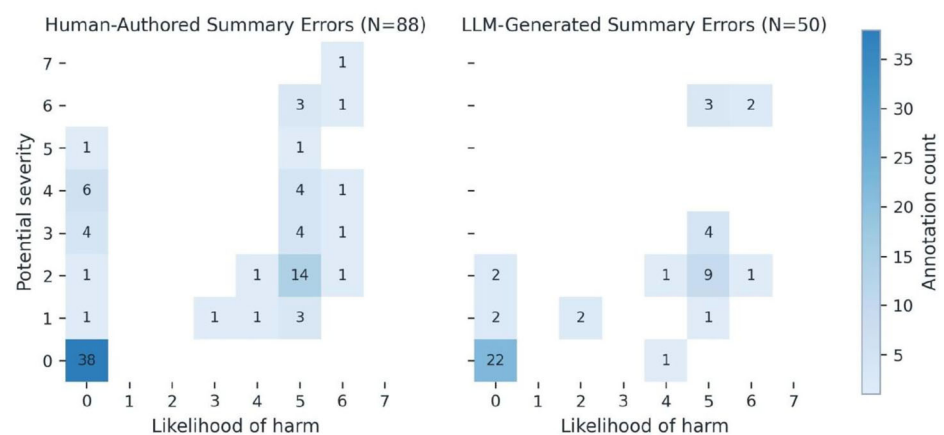
For most encounters (57/60, 95%), the physician reviewers preferred the LLM-generated summary to the human-authored summary. For the remaining 5% (3/60 encounters), the human-authored summary was preferred, and the “no preference” option was never selected. The mean scores on the physician reviewer surveys were higher for LLM-generated than human-authored discharge summaries across all metrics: quality (mean [SD] 4.71 [0.47] vs 3.16 [1.08]; $p = 4.28\text{e-}6$), readability (4.66 [0.56] vs 3.39 [1.04], $p = 1.93\text{e-}4$), factuality (4.62 [0.62] vs 3.73 [0.85], $p = 1.09\text{e-}5$), and completeness (4.69 [0.53] vs 3.42 [1.11], $p = 2.92\text{e-}5$). PCP reviewers rated LLM-generated summaries as easier to understand than human-authored summaries (4.75 [0.44] vs 3.20 [1.12], $p = 0.009$), easier to communicate to patients and caregivers (4.75 [0.47] vs 3.27 [1.12], $p = 0.011$), and easier to use for providing follow-up care (4.68 [0.57] vs 3.20 [1.19], $p = 0.013$) than the human-authored summaries. The mean Likert scores, or the average score for each question, for physician reviewer survey questions are shown in Table 1.

Physician reviewer ratings were more aligned for LLM-generated summaries than for human-authored summaries. The mean absolute deviation in Likert scores for the LLM-generated summaries was less than half that of the human-authored summaries (0.49 vs 1.07, $p = 0.001$). Similarly, the exact match percentage, or proportion of paired survey responses where both reviewers assigned the same Likert score, for the LLM-generated summaries was more than double that of the human-authored summaries (57% vs 27%), and only 4% of the paired LLM-generated summary ratings had a difference of more than 1 point, versus 27% for the human-authored summaries.

Across 60 encounters containing 138 error annotations (47 inaccuracies and 91 omissions), there were fewer annotated errors per LLM-generated summary than per human-authored summary (mean count [SD] 0.42 [0.98] vs 0.73 [1.34], $p = 0.011$). Mean potential and likelihood of harm score distributions between LLM-generated and human-authored summaries were similar (1.46 [1.84] vs 1.73 [1.89], $p = 0.380$ for harm potential; 2.30 [2.51] vs 2.11 [2.52], $p = 0.648$ for harm likelihood). Within error categories, there was a lower rate of omissions with LLM-generated vs. human-authored summaries, but no difference in the rate of inaccuracies (0.26 [0.72] vs. 0.5 [1.0], $p = 0.004$ for omissions; 0.16 [0.54] vs 0.23 [0.72], $p = 0.45$ for inaccuracies). Figure 1 shows the annotated error counts and distributions of harm scores for both human-authored and LLM-generated discharge summaries.

Table 1 | Mean survey results of physician reviewer ratings for human-authored vs. LLM-generated discharge summaries, based on a 1-5 Likert rating scale

Question (n)	Evaluators	Human-authored summary mean Likert score	LLM-generated summary mean Likert score	Mean score difference between LLM-generated & human-authored summaries	P-value
Overall quality (n = 120)	Hospitalists, PCPs	3.16	4.71	1.55	4.28E-6
Concise/readable (n = 120)	Hospitalists, PCPs	3.39	4.66	1.27	1.93E-4
Factuality (n = 120)	Hospitalists, PCPs	3.73	4.62	0.89	1.09E-5
Completeness (n = 120)	Hospitalists, PCPs	3.42	4.69	1.27	2.92E-5
Easy to understand (n = 60)	PCPs	3.20	4.75	1.55	0.009
Easy to verbalize (n = 60)	PCPs	3.27	4.75	1.48	0.011
Easy to follow up (n = 60)	PCPs	3.20	4.68	1.48	0.013

Fig. 1 | Heat map of physician reviewer-annotated error counts and harm ratings by summary type. Each cell counts error annotations with the corresponding potential-severity and harm-likelihood ratings on the 0–7 scales.

For both human-authored and LLM-generated summaries, length of stay was not significantly correlated with mean Likert score (averaged across the four questions answered by both hospitalists and PCPs) for either summary type (Spearman $\rho = 0.027$, $p = 0.837$ for human-authored summaries; 0.138 , $p = 0.292$ for LLM-generated summaries). Figure 2 shows the relationship between mean survey ratings and length of stay for both human-authored and LLM-generated discharge summaries.

Clinician perspectives on summary quality and usability

Themes identified from the debriefing sessions are shown in Supplementary Table 4. During the debriefing sessions, physician reviewers shared their insights comparing AI (LLM)-generated and human-authored summaries. Qualitative data concurred with the quantitative results. Specifically, PCP and hospitalists mentioned the following as main reasons underlying a preference for LLM-generated summaries: reduction of cognitive burden, higher consistency of summary qualities, and clearer presentation of critical information such as treatment rationale and unresolved follow-up issues. Most physicians found LLM-generated summaries lightened their cognitive burden in reading discharge summaries. One PCP noted that “AI summaries were just, obviously, I felt more understandable, more succinct, better organized. It told the story. It was easier to read rather than just kind of piecing through it and trying to understand.” Whereas for human-authored discharges, physicians sometimes “spend like an hour or two ... scouring the chart, trying to figure out what happened ... and can’t even figure out what happened”.

Physician reviewers noted consistency in the quality of LLM-generated summaries. One PCP mentioned that for human-authored summaries, they are “physician-dependent” such that “some were better than others...some were quite detailed while others were just maybe three paragraphs. For LLM-generated summaries, they were “more uniform...more understandable, more succinct, better organized.” Another hospitalist concurred with the PCP by noting that “[the main difference] is the inconsistency between physicians... [for human-authored summaries] there can be redundancy... there are so many things that are omitted that are important...Whereas the AI summaries are concise and there was no redundancy”.

Another reason for physician reviewers’ high ratings for LLM-generated summaries concerned the clarity of critical health information. Physician reviewers mentioned that treatment rationale was key information that they looked for in the discharge summaries. One PCP mentioned an instance where the LLM-generated summary included treatment rationale, whereas the corresponding human-generated summary lacked this information. Physicians also mentioned unresolved follow-up issues as key discharge summary information. One hospitalist noted that LLM-generated summaries did a better job presenting information regarding unresolved follow-up issues: “what I loved [about LLM-generated summaries], is that anything that was still pending was acknowledged and had a comment on who is appropriate to follow up with [for] these results...that is something that was essentially not found in any of the human-written summaries”.

Fig. 2 | Scatterplot comparing mean survey ratings to length of stay for both human-authored and LLM-generated discharge summaries ($n = 60$). Individual values of n above each data point indicate the number of encounters available to calculate the average.

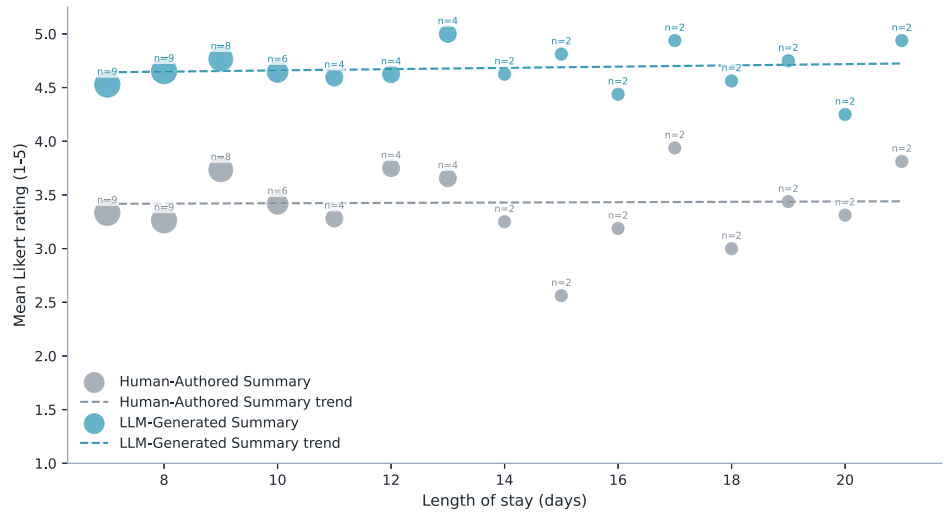
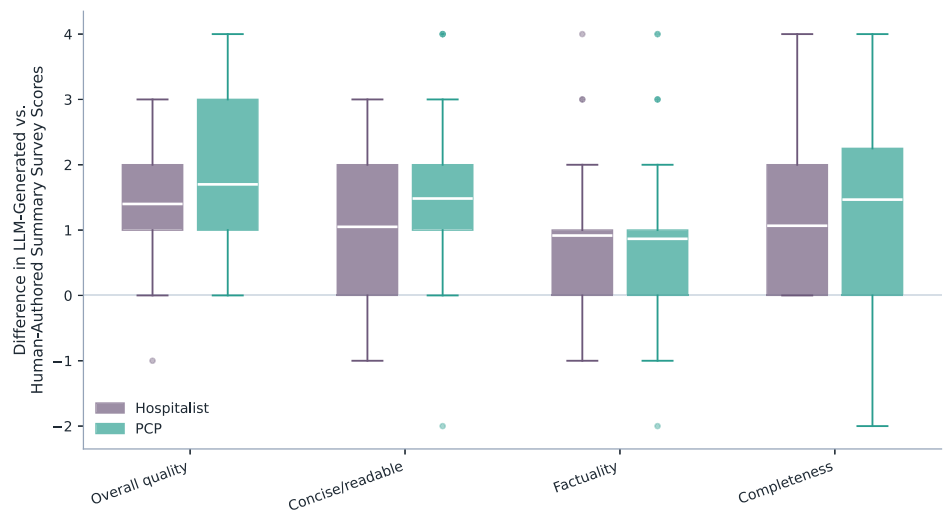


Fig. 3 | Difference in LLM-generated vs. human-authored survey scores by provider type and survey question ($n = 120$). Outliers are shown as individual points but excluded from mean, IQR and range calculations.



Provider-specific differences in discharge summary evaluation

The difference in scores between LLM-generated and human-authored summaries by provider type and survey question is shown in Fig. 3. The mean score differences were higher for PCPs than hospitalists for readability/conciseness and completeness (mean [SD] 1.48 [1.28] vs 1.05 [1.14], $p = 0.047$ for readability/conciseness; 1.47 [1.38] vs 1.07 [0.97], $p = 0.044$ for completeness; Supplementary Table 6). Between providers, mean harm score distributions were similar for harm potential (mean [SD] 1.65 [1.83] vs 1.58 [2.02], $p = 0.675$), but PCPs assigned a higher likelihood of harm than hospitalists (2.66 [2.51] vs 0.83 [1.98], $p = 5.33 \times 10^{-4}$). PCPs and hospitalists annotated similar counts of inaccuracies per summary (mean [SD] 0.40 [0.827] vs 0.38 [1.03], $p = 0.664$), but PCPs annotated more omissions than hospitalists (1.3 [1.84] vs 0.22 [0.585], $p = 1.16 \times 10^{-5}$). Overall, hospitalists and PCPs annotated similar numbers of inaccuracies (49% vs. 51%), but PCPs annotated more omissions (86% vs. 14%).

Provider perspectives on post-discharge risk and clinical workflows

To better understand the differences in ratings between PCPs and hospitalists, we probed physician reviewers' own interpretations during the debriefing sessions. PCP reviewers clearly attributed higher harm likelihood scores to their long-term experience and view of patients. They explained that the reason why PCPs rated harm likelihood much higher was that they had experienced and had to deal with many medical complications after discharge. One PCP shared that "... because we've experienced, you know,

somebody sent home on 2 anticoagulants...blood thinners...excessive psychiatric medications, too much diuretic. We see what happens. And often we're the ones that have to deal with the complications." There were also differences noted in risk tolerance between inpatient and outpatient care settings. One hospitalist noted that "(PCPs) throw the dice every day... So they probably have seen a lot more stuff than I've seen... So, like, they've seen a lot more of, like, bad stuff in non-controlled settings."

Differences in provider workflows also emerged as an important theme. For PCPs, they described their workflow in reviewing discharge summaries for transition of care and the need to request records and review details within individual EHR notes when discharge summaries were insufficient. One PCP shared that "I read the discharge summary, the medication reconciliation in there, and then I more often than not don't trust it and go back and I look at vital signs, labs, micro and imaging from the hospitalization, go over it myself." From the hospitalist's perspective, the focus was on reviewing and synthesizing historically important information from the EHR. As a hospitalist noted, "I know I took over a patient once who had been in the hospital for over 3 months without a good hospital course. And it took me 2 h to go through that chart, just to generate a discharge summary."

Evaluation of incidental finding identification

There were 31 LLM-identified incidental findings from radiology reports in the LLM workflow. Physician reviewers rated 93.5% of the findings as

factually correct and 87.1% appropriate to be reported as an incidental finding on the discharge summary (Supplementary Table 7).

Some notable differences in perspectives on incidental finding reporting emerged during debriefing sessions among PCPs and hospitalists. Several hospitalist reviewers expressed their favor for a broader inclusion of incidental findings. One noted that “As a hospitalist, I am covering my own bases, right? So I’m calling out everything (that could be an incidental finding), and then I let (the PCP) decide which ones he wants to follow or not.”

Discussion

LLM-Generated Summaries Improve Quality and Reduce Cognitive Burden. In this study of patients with hospitalizations between 7 and 21 days, our results suggest that LLM-generated discharge summaries outperformed human-authored summaries in producing more understandable and actionable handoffs without introducing substantial errors. Across evaluations by both hospitalists and PCPs, LLM-generated summaries received higher ratings for overall quality, readability/conciseness, factuality, and completeness, and physicians indicated a preference for the LLM-generated summary in 95% (57/60) of encounters. Importantly, PCPs, the intended recipients of discharge summaries for transition-of-care management, rated LLM-generated summaries higher for ease of understanding and verbalizing the major aspects of the hospital course to patients, and for supporting follow-up care. Feedback from the debriefing sessions suggests that these differences were meaningful because LLM-generated summaries reduced the cognitive burden of reconstructing chronological events, improved the consistency in quality across discharge summaries, highlighted rationale for treatment changes, and improved identification of unresolved issues requiring follow-up. These findings build on prior studies demonstrating that LLM-generated discharge summaries achieve quality comparable to or better than physician-authored summaries by showing that these benefits extend to longer, more clinically complex hospitalizations. Furthermore, our qualitative findings complement the prospective work of Grolleau et al. by suggesting that reductions in cognitive burden may represent an important mechanism through which LLM-generated summaries improve clinician experience, even when time savings are modest¹¹.

Safety and Error Characteristics of LLM-Generated Summaries. Analysis of physician-annotated errors demonstrated that human-authored summaries contained more annotated errors than LLM-generated summaries, with omissions representing the most common type in both groups. Annotated errors in human-authored summaries more often involved omission of clinically important details related to hospital management and transition-of-care planning, including treatments, medication discontinuation plans, relevant consultations and follow-up recommendations. In contrast, physician-annotated errors in LLM-generated summaries were generally narrower in scope and more commonly involved isolated omissions of medication details, follow-up timing, or contextual information such as relevant comorbidities prior to hospitalization. Importantly, there was no difference in the potential severity or likelihood of harm between annotated errors in LLM-generated compared to human-authored summaries. These reviewer annotations suggest that while LLM-generated summaries are not error-free and require physician oversight, reviewers did not identify more clinically meaningful omissions or inaccuracies than those that commonly occur in human-authored discharge summaries.

LLM Performance for Long and Clinically Complex Hospitalizations. The stability of findings across 7- to 21-day hospitalizations has important implications, with LLM-generated summaries maintaining consistently superior performance despite increasing information volume. Longer encounters often produce larger quantities of notes, repetitive information, and complex clinical trajectories that are difficult for both discharging and receiving physicians to synthesize. Debriefing sessions suggest that these differences may translate into meaningful time savings and reduced cognitive burden in reconstructing the hospital course from fragmented documentation. Participants noted that with longer hospitalizations, time constraints often force clinicians to skim lengthy records, increasing the risk

that important issues such as the rationale for management decisions or unresolved follow-up issues may be omitted. In addition, the methods used in this study demonstrate that LLMs can generate coherent discharge summaries even for EHRs exceeding the model’s context window, an important technical consideration given the large volume of documentation associated with longer hospitalizations. Whereas previous studies largely evaluated hospitalizations of approximately 2–8 days, our findings suggest that LLM-generated summaries maintain high performance despite the substantially greater documentation burden associated with prolonged hospitalizations, addressing an important gap in the current literature. As LLM context windows continue to expand, future models may be able to process entire hospital encounters in a single pass, reducing the need for multi-step summarization pipelines. However, larger context windows alone may not eliminate the need for methods that prioritize, synthesize, and organize clinically relevant information from lengthy hospitalizations, suggesting that the principles underlying our approach may remain relevant as LLM capabilities evolve. Additionally, EHR documentation artifacts, such as dot phrases and carried-forward text, may influence summary quality and warrant further study in LLM-based summarization workflows.

Differences Between PCP and Hospitalist Perspectives. The advantages of LLM-generated summaries appeared to be more pronounced among PCPs than hospitalists. Across both overall and paired evaluations, PCPs demonstrated larger differences in ratings favoring LLM-generated over human-authored summaries, particularly for overall quality and conciseness/readability, with less variability between PCP and hospitalist ratings for the LLM-generated summaries. PCPs also annotated more errors, specifically omissions, in both human-authored and LLM-generated summaries than hospitalists, with higher scores for likelihood of harm. Debriefing sessions suggested several reasons for this difference. PCPs often perceive greater downstream harm potential from omissions or inaccuracies because they are responsible for longitudinal follow-up and management of post-discharge complications. As one PCP noted, “we see what happens, and often we’re the ones that have to deal with the complications,” contributing to a higher perceived likelihood of medication or follow-up errors from incomplete discharge documentation. Additionally, workflow differences likely contributed to the discrepancy in omission annotation, as PCPs relied on discharge summaries to reconstruct the hospitalization and often reviewed inpatient notes when key details were missing, whereas hospitalists could draw on their contextual knowledge of the inpatient course, making omissions less apparent. These findings also highlight that judgments regarding omissions are influenced by reviewer role and workflow, underscoring that discharge summary completeness is not a single objective construct but depends on the informational needs of the intended user. PCPs may perceive greater value from LLM-generated summaries because their workflow requires directly acting on discharge documentation to provide ongoing care, whereas hospitalists are primarily focused on creating a summary for handoff. The prompt engineering in this study was designed with PCPs as the target audience, suggesting that incorporating PCP informational priorities and workflow needs into development may improve the quality and usability of discharge summaries for receiving providers during the transition of care. More broadly, our findings suggest that a one-size-fits-all discharge summary may not optimally serve all stakeholder needs, and that LLM systems have the potential to tailor summaries to wider audiences, including hospitalists, PCPs, and patients.

Incidental Finding Identification and Reporting. Our results suggest that LLMs performed well as a tool for identifying and summarizing incidental findings. Most incidental findings identified in LLM-generated summaries were judged to be both appropriate and correctly reported, and notably, reviewers did not annotate omission errors related to incidental findings from radiology studies. Debriefing sessions highlighted differences between hospitalist and PCP perspectives on incidental findings, with hospitalists favoring broader reporting to ensure abnormalities were documented and questioning whether LLMs might underreport incidental findings, while PCPs preferred selective reporting of actionable findings to avoid information overload. Importantly, the prompt for incidental findings

was designed for specificity, where the LLM was directed to identify findings that were specifically flagged by radiologists for follow-up. However, both groups agreed that LLM-generated incidental finding summaries were beneficial for reducing the likelihood that clinically important findings requiring follow-up would be missed. Although prior studies have demonstrated that LLMs can accurately identify incidental findings from radiology reports, our study extends this work by evaluating incidental finding extraction within the context of discharge summary generation, where balancing completeness with information overload is critical for the transition of care.

Implications for Clinical Adoption and Oversight. Although both quantitative evaluation and qualitative debriefing indicated the tremendous potential of LLM-generated summaries, physicians cautioned against blind reliance on LLMs. When probed with final thoughts on adopting LLM-generated summaries in clinical workflows, physicians emphasized that LLM summaries should only function within a clinician-supervised context. Physicians in all three debriefing sessions strongly agreed that, as reflected in one hospitalist's quote, "no matter what information is pre-populated or anything, it's still the responsibility of the physician to make sure it's correct." Physicians specifically highlighted the importance of deliberate verification workflows for high-risk content such as medication changes, follow-up recommendations, and incidental findings. Moreover, some had cautionary comments on potential threats to the widespread adoption of LLM-generated summaries in clinical workflows. One hospitalist cautioned that "I guarantee you no one's going back and fact-checking or reading everything to make sure that it's fine... Because usually when something's handed to you, you're just going to take it as is and you give up your agency to kind of do some of the things that doctors do... I'm afraid that, you know, they're not going to do any QA [quality assurance] to make sure that everything is correct." Together, these findings suggest that the greatest value of LLM-generated discharge summaries lies not simply in automating documentation, but in improving the quality, consistency, and usability of transition-of-care communication while supporting clinician oversight.

This study has several limitations. First, it was conducted at a single academic medical center using one EHR environment and local documentation practices, and the input pipeline included only clinically relevant notes selected by a single physician informaticist, which may introduce selection bias and limit generalizability to other health systems, specialties, or workflows. Second, because reviewers were not blinded to the source of the summaries, expectation bias cannot be excluded; however, the historical and LLM-generated summaries differed substantially in format, making blinding impractical without modifying the original clinical documents and reducing the real-world nature of the comparison. Third, since reviewers evaluated summaries using a restricted view of clinical notes rather than the full production EHR, findings may not fully reflect routine clinical use and should be confirmed in future prospective EHR-integrated evaluations. Fourth, the study evaluated physicians' perceptions of discharge summary quality and safety rather than downstream patient-centered outcomes such as timely completion of discharge summaries, readmissions, medication errors, or adverse events after discharge. Although clinician assessment is highly relevant for transition-of-care workflows, future prospective studies are needed to determine whether these perceived advantages translate into measurable improvements in patient outcomes and operational efficiency. Fifth, because each encounter was reviewed by two reviewers of different specialties, the design does not permit estimation of formal measurement reliability. Future studies could have repeated ratings by multiple reviewers within the same rater population or other designs that allow measurement-reliability calculation. Sixth, because omissions and inaccuracies were annotated by one hospitalist and one PCP per encounter without within-specialty adjudication or a consensus reference standard, these findings should be interpreted as clinician judgments rather than objective measures of error. Additionally, we did not characterize systematic versus idiosyncratic error patterns, as our annotation framework was not designed for this purpose. Future studies should add adjudication or a reference standard for errors and evaluate whether LLM-generated summaries exhibit recurring

blind spots that warrant targeted verification. Furthermore, the evaluated pipeline was frozen before the formal clinician review, and reviewer-identified errors were not used to refine prompts, regenerate outputs, or retrain the model. Because the aim of this study was to evaluate clinician perceptions of a fixed system's output rather than to optimize the system itself, this avoided adapting the pipeline to the evaluation cases; however, it also means the study does not assess whether iterative, error-driven refinement could reduce recurring errors. In operational deployment, continuous monitoring and error-driven improvement would be essential, and we identify prompt refinement, model updating, and prospective re-evaluation on an independent cohort as important next steps. We did not evaluate information contained in documents created after the final progress note, and the impact of this cutoff on omitted content warrants future study. Lastly, while qualitative debriefing sessions provided important contextual insights into workflow, trust, and clinician perceptions, we did not use formal qualitative research methodologies such as structured thematic coding or saturation analysis, and these findings should therefore be interpreted as exploratory.

Conclusion. In conclusion, LLM-generated discharge summaries for longer hospitalizations were rated higher than human-authored summaries across multiple performance metrics by both hospitalists and PCPs. The perceived benefits and quality advantages were particularly pronounced among PCPs, suggesting that designing discharge summaries around the informational priorities and workflow needs of receiving providers may meaningfully improve transition-of-care communication for those who provide post-hospitalization care. These findings support the potential for thoughtfully implemented LLM-assisted summarization to reduce cognitive burden and better support transitions of care, while maintaining the need for clinician oversight and verification.

Methods

Study cohort

We extracted clinical notes, imaging reports, and discharge summaries from 400 historical inpatient encounters on the Internal Medicine service at Stony Brook University Hospital between January 1, 2023 and December 31, 2024, with lengths of stay ranging from 7 to 21 days. Based on the availability of resources and time needed for clinician review, we analyzed 60 encounters; the initial sample size of 400 encounters was chosen as a sufficient buffer of extra data to allow for the creation of a held-out development dataset as well as the exclusion of peculiar encounters unrepresentative of the typical distribution. Encounters without discharge summaries or with fewer than 25 clinical notes were excluded after manual analysis revealed encounters with fewer than ~25 clinical notes to be largely idiosyncratic. For each encounter, we extracted all clinical notes from the EHR, which included 654 unique note types with textual information across multiple categories, including emergency department notes, consult notes, admission and history and physical notes, progress notes, care management notes, addenda, and event notes. Note types were reviewed by a physician informaticist and filtered for inclusion as LLM inputs based on clinical relevance for discharge summary generation. All data storage and computation were performed via a secure shell on an in-network computing cluster. Inputs to GPT-5.2 were submitted through an API to a private HIPAA-compliant Microsoft Azure OpenAI Service instance. GPT-5.2 was selected for its frontier performance (at the time of data collection) and preferred over Anthropic models, the other frontier-performing model provider, due to limitations in our institution's business associate agreement, which restricted private health information from being used with Anthropic models. Self-deployment of high-performing open-weight models was considered, but available computational resources would have limited us to smaller models with substantially lower expected performance than frontier models.

The study was approved by the Stony Brook University Office of Research Compliance (Institutional Review Board) under protocol IRB2024-00457. The study was approved under exemption category 45 CFR 46.104(d)(4)(iii). The requirement for informed consent was waived by the Stony Brook University Institutional Review Board in accordance with 45

Table 2 | Scores and definitions of potential and likelihood of harm included in the instructions for physician reviewer error annotation

Harm scores	Harm potential definition	Harm likelihood definition
0 (None)	No potential for harm	Impossible (0 in 10 M)
1 (Minimal)	Potential for emotional distress or inconvenience (mild and transient anxiety or pain or physical discomfort)	Nearly Impossible (≤ 1 in 10 M)
2 (Minor)	Potential for requiring additional treatment	Extremely Rare (1 in 1 M)
3 (Moderate)	Potential for temporary harm (bodily or psychological injury, but likely not permanent)	Very Rare (1 in 100 K)
4 (Significant)	Potential for permanent harm (lifelong bodily or psychological injury or increased susceptibility to disease)	Rare (1 in 10 K)
5 (Serious)	Potential for lifelong bodily or psychological injury or disfigurement	Possible (1 in 1000)
6 (Severe)	Potential for severe permanent harm	Likely (1 in 100)
7 (Catastrophic)	Potential for death	Near Certain (>1 in 10)

CFR 46.116(f), as this study involved retrospective secondary analysis of existing clinical data and posed minimal risk to participants. The research was conducted in accordance with applicable federal regulations, institutional guidelines, and the ethical principles of the Declaration of Helsinki.

Encounter processing

Notes for each encounter extracted from the Oracle Cerner EHR were provided in rich text format (RTF) and converted to plain text for LLM processing using the Python stripf library (Python 3.12). Plain-text notes and encounter-level metadata were stored in a SQLite database using sqlite3 (Python 3.12). Notes were ordered chronologically, and those published after the final progress note for the encounter were excluded to prevent leakage of information unavailable at the time of discharge. Notes containing fewer than 40 characters were also removed, as these were primarily attestation entries. For each encounter, the historical human-authored discharge summaries were extracted for comparison with the LLM-generated summaries.

LLM discharge summary generation

A flow diagram of the technical steps in LLM discharge summary generation is shown in Supplementary Figure 1. Because GPT-5.2 has a context limit and longer encounters could not be processed in a single pass, we divided the clinical documents into 50,000-token segments and inputs were assembled in a SQLite database. The LLM was prompted to create a chronological, bulleted draft of key components of the discharge summary (Supplementary Fig. 2) using iterative refinement with a sliding window over the note segments. The model generated an initial summary from the first segment of the record and then updated the summary as successive segments were processed, ensuring that each call remained within the window limit. We used a window size of 50,000 tokens to balance token costs, latency and output quality. In parallel, a separate prompt was developed for identification and extraction of incidental findings from radiology reports (Supplementary Fig. 3). The LLM was then prompted to consolidate the chronological, bulleted draft and incidental findings into a final narrative summary. Pipeline development utilized LangChain and LangGraph in Python 3.12.

Discharge summary evaluation

Evaluation of both human-authored and LLM-generated discharge summaries was conducted with reviews by 12 attending physicians (6 hospitalists and 6 PCPs with representation from inpatient, outpatient, and informatics leadership) using quantitative evaluation followed by focus group debriefing sessions. A stratified random sample of 60 encounters was selected to match the length-of-stay distribution of the full dataset. Each physician was assigned to review 10 encounters, with every encounter independently evaluated by one hospitalist and one PCP. This design was chosen to balance the need for independent assessments from two clinically relevant perspectives with the substantial time and cognitive effort required to review an encounter. Requiring multiple evaluations across several

encounters would have increased reviewers' workload and substantially reduced the feasibility of recruiting experienced (and busy) clinicians to complete the evaluations. We therefore prioritized broad coverage of encounters by reviewers with different specialties over a limited number of repeated within-specialty ratings. Reviewers were not blinded because the historical and LLM-generated summaries differed substantially in format, and preserving the original documents provided a more clinically realistic comparison than an artificially reformatted control. To align reviews with real-world clinical workflows, hospitalists were instructed to review all inpatient clinical notes prior to evaluation of the human-authored and LLM-generated discharge summaries, and PCPs were instructed to first review the discharge summaries and then corroborate findings with the inpatient clinical notes (Supplementary Fig. 6). Reviewer training was conducted via virtual meetings prior to evaluation and covered use of the secure review application, definitions and examples for each Likert domain, identification of inaccuracies and omissions, application of the adapted AHRQ potential- and likelihood-of-harm scales, incidental-finding questions, role-specific review sequence, and procedures for corroborating annotations using source notes. Reviewers were allowed to review either summary in their own choice of order and compare the summaries throughout the review for a more ecologically realistic setting based on prior reviewer experiences.

Reviews were conducted via a custom web application deployed behind the University Hospital firewall; Supplementary Fig. 5 shows example screenshots demonstrating the main features of the web application. Evaluators were provided access to review the information through tabs presented left-to-right with (1) all clinical notes that were used as inputs, (2) the human-authored summaries and evaluation survey, (3) the LLM-generated summary and evaluation survey, (4) LLM-identified incidental findings, (5) evaluation of overall preferences. The user interface also included a keyword search functionality for locating relevant information within the EHR documentation and features for completing survey questions, annotating and scoring errors within the human-authored and LLM-generated discharge summaries. Reviewers assessed summaries using the same pre-processed clinical notes and imaging reports provided to the LLM to enable standardized evaluation and source verification within a single evaluation platform.

The evaluation for all reviewers included four questions with a 5-point Likert scale to assess overall quality, conciseness/readability, factuality and completeness⁷ as well as preference for the human-authored versus LLM-generated summary. PCPs were assigned three additional survey questions on ease of understanding, verbalizing to patients and caregivers, and following up on relevant aspects of the hospitalization (Supplementary Table 1). To evaluate discharge summary errors, reviewers were instructed to annotate errors by type (inaccuracies and omissions) for both human-authored and LLM-generated discharge summaries (Supplementary Table 2). For each error annotation, reviewers provided a brief description and rated the likelihood and potential for patient harm using an adapted Agency for Healthcare Research and Quality harm scale (Table 2)^{8,12}. For

encounters where the LLM detected clinically relevant incidental findings, reviewers were presented with the finding and asked to rate whether it was reported accurately and whether they thought reporting was appropriate (Supplementary Table 3). We did not evaluate incidental findings on human-authored discharge summaries as there was variability in documentation of them across historical summaries.

Three virtual, post-review, semi-structured debriefing sessions were held with physician evaluators to discuss findings from the surveys and error annotations and to obtain qualitative feedback on their perspectives. Sessions were recorded and transcribed verbatim; we held three sessions to allow for maximum participation so that each physician could attend a debrief session. We analyzed the resulting transcripts with a hybrid approach that combined human coding with LLM assistance. This involved the use of GPT 5.5, with reasoning effort set to high, to analyze the meeting transcripts and cluster related answers to our questions as well as pair converging/diverging opinions of PCPs and hospitalists on shared topics across the three sessions. The LLM then outputted an interactive website with the analysis superimposed over the transcripts and links between utterances and synthesis. We then analyzed each LLM-posed insight against our prior, manual analysis of the debrief session transcripts, in which we performed similar tasks as given to the LLM, to identify additional items for discussion, discarding those that did not verifiably enhance the manual analysis.

Statistics and reproducibility

For each survey question, the difference between the LLM-generated and human-authored summary ratings was calculated for every reviewer-encounter evaluation. Mean rating differences were analyzed using two-sided intercept-only linear models with two-way cluster-robust standard errors clustered by reviewer and encounter. *P* values were calculated using *t* distributions with degrees of freedom equal to the smaller number of clusters minus one: 11 degrees of freedom for questions answered by all reviewers and 5 degrees of freedom for PCP-only questions. Differences in LLM-versus-human rating changes between hospitalists and PCPs were compared using two-sided paired Wilcoxon signed-rank tests at the encounter level. Error annotation counts were compared between summary types and between provider types, both overall and by error type, using two-sided paired Wilcoxon signed-rank tests. Two-sided Mann–Whitney *U* tests were used to compare the harm potential and harm likelihood score distributions for annotated errors in human-authored versus LLM-generated summaries and by provider type. Error annotations differed in number and content between summary types and could not be matched one-to-one; therefore, paired testing was not applicable to the harm-score distributions. Associations between length of stay and the mean Likert scores, or the average of the scores for the 4 physician reviewer questions, were assessed using two-sided Spearman rank correlation tests, separately for human-authored and LLM-generated summaries. Nonparametric tests were used for provider comparisons, annotation counts, harm scores, and length-of-stay associations because these variables were not assumed to be normally distributed. For all tests, *p* < 0.05 was considered statistically significant.

We used the 2025-12-11 snapshot of GPT 5.2 via Microsoft Azure Foundry. Outputs were generated via a single sample in March 2026. Traditional model hyperparameters, including temperature, top-*p* and seed, could not be set, and their internal values were not observable because we set the reasoning effort hyperparameter to medium. Traditional hyperparameters for GPT 5.2 can only be set/observed if reasoning effort is set to none, which often degrades output quality. The maximum number of output tokens was set to the default setting, meaning that the model selected the maximum number of output tokens dynamically.

The full prompts used to generate model outputs in this study are available in Supplementary Information Figs. S2, S3 and S4.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to containing private health information (PHI) but are available from the corresponding author on reasonable request.

Code availability

The code and raw, de-identified results used to generate the statistical analyses are publicly available at <https://github.com/StonyBrookDB/Clinician-Centered-LLM-Discharge-Summaries>.

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Author contributions

T.O. and S.J. performed all data analysis and pipeline design for this study, as well as software engineering to build the web interface. T.M. provided clinical expertise with prompt engineering and review of outputs. X.Z. assisted with the qualitative aspects of study design, data collection, and manuscript writing. S.A., S.H., K.K., S.L., P.N., K.O., L.S., R.S., T.S., M.T., Z.W. and R.W. provided hospitalist and primary care expertise, reviewed

discharge summaries and contributed clinical and workflow insights during debriefing sessions. R.W. conceived of the study and made most methodological decisions with the assistance of T.O. and F.W. T.O., R.W. and X.Z. wrote the manuscript, with edits and review by all coauthors. Declaration

Competing interests

The authors declare no competing interests.

Additional information

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